

ECE 251A: Programming Assignment

Yinyan Bu

4/28/2026

Goals

- ▶ This Numerical Assignment has two components:
- ▶ Classical PSD estimation:
 - ▶ Periodogram
 - ▶ Welch method
 - ▶ Statistical behavior analysis
- ▶ Machine learning based spectral estimation:
 - ▶ Train simple MLP/LSTM to learn stochastic process generation
 - ▶ Compare learning-based and classical approaches
 - ▶ Estimate system impulse response from learned outputs

Allowed Functions vs What You Expect to Write

You are allowed to use:

- ▶ Basic linear algebra and signal processing functions like
 - ▶ filter functions
 - ▶ window functions
 - ▶ convolution
 - ▶ FFT
- ▶ Statistical random number generation functions
- ▶ Any plotting libraries and functions (e.g., Matplotlib)

You are expected to write/modify yourself:

- ▶ Neural Model configuration exploration
- ▶ Hyperparameter exploration
- ▶ Visualization/Analysis codes
- ▶ Data preparation codes
- ▶ Format transfer codes (potential)

Part I: Periodogram Estimation

- ▶ (a): make sure to compute autocorrelation over some reasonably long period of time (i.e., reasonably many time samples of $h[n]$).
- ▶ (b): first generate $x[n]$ and then use $x[n]$ and $h[n]$ to get $y[n]$.
- ▶ (c): the true power spectrum of the system is given by $|H(e^{j\omega})|^2$, think about how to use $h[n]$ to get $H(e^{j\omega})$.
- ▶ (d): implement periodogram estimation using FFT of $y[n]$.
- ▶ (e): first generate a large number (you can start with hundreds or thousands) of sequences of $y[n]$ and do what you did in (d) for each realization. Then average over these trials.
- ▶ (f): repeat (e) using Welch periodogram estimation with your own choice of window function.

Part II: Deep Learning Based Power Spectrum Estimation

- ▶ Download all the codes into a folder, install all required libraries.
- ▶ Prepare your own data and put them in the same folder.
- ▶ (a): play the training of models with different configurations/parameters, get yourself familiar with implementation of codes.
- ▶ (b): play the training of models with different value of P , you can keep other configurations/parameters fixed.
- ▶ (c): generate the desired number of realizations of $x[n]$ → save them in one file → call the infer function to infer all the realizations of $y_{ML}[n]$ in one time → repeat 1(e) using the realizations of $y_{ML}[n]$.
- ▶ (d): write down how would you mathematically estimate the impulse response using sample correlation of $x[n]$ and $y_{ML}[n]$, then implement it and compare it with the ground truth.